Q:1 Explain about the Learning outcomes for studying the Computer Organization and Architecture course ?

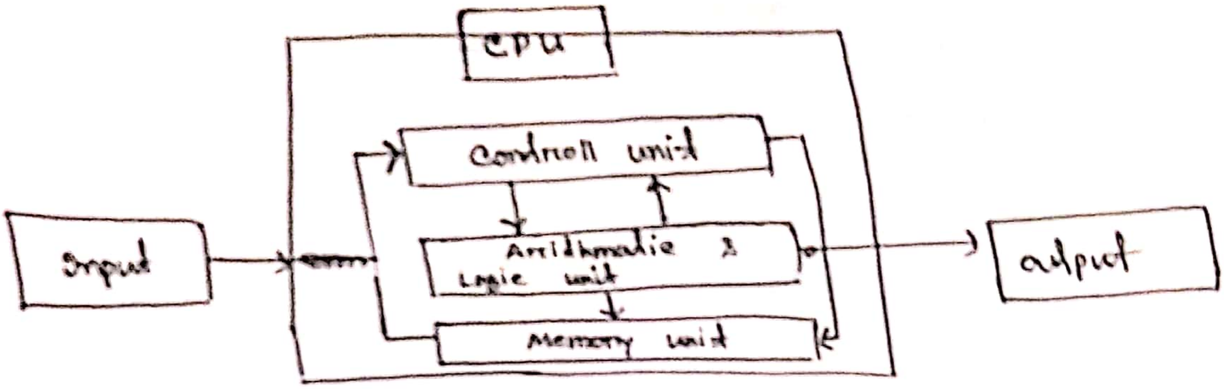
Answer:- Its impossible to get well-round computer science education without understanding computer organization and architecture because too many things depend on such understanding.

* performance analysis of practical software
* Embeded and mobile computing
* High performance game programming
* High performance database.
* Accelerators, GPU computing and related topic
* Compilers and code optimization

omputer organization and architecture lets you know how xactly each instruction is executed at the micro level. f you study processor design you must need to know mputer organization and architecture. Assembly Langua ll help you to know exactly how many instructions-cycle ll take to execute a code. Memory hierarchy odel help you optimize file / cache / RAM.

(11) Illustrate the basic elements of digital Computer ?

⇒ Major components of digital computer :-

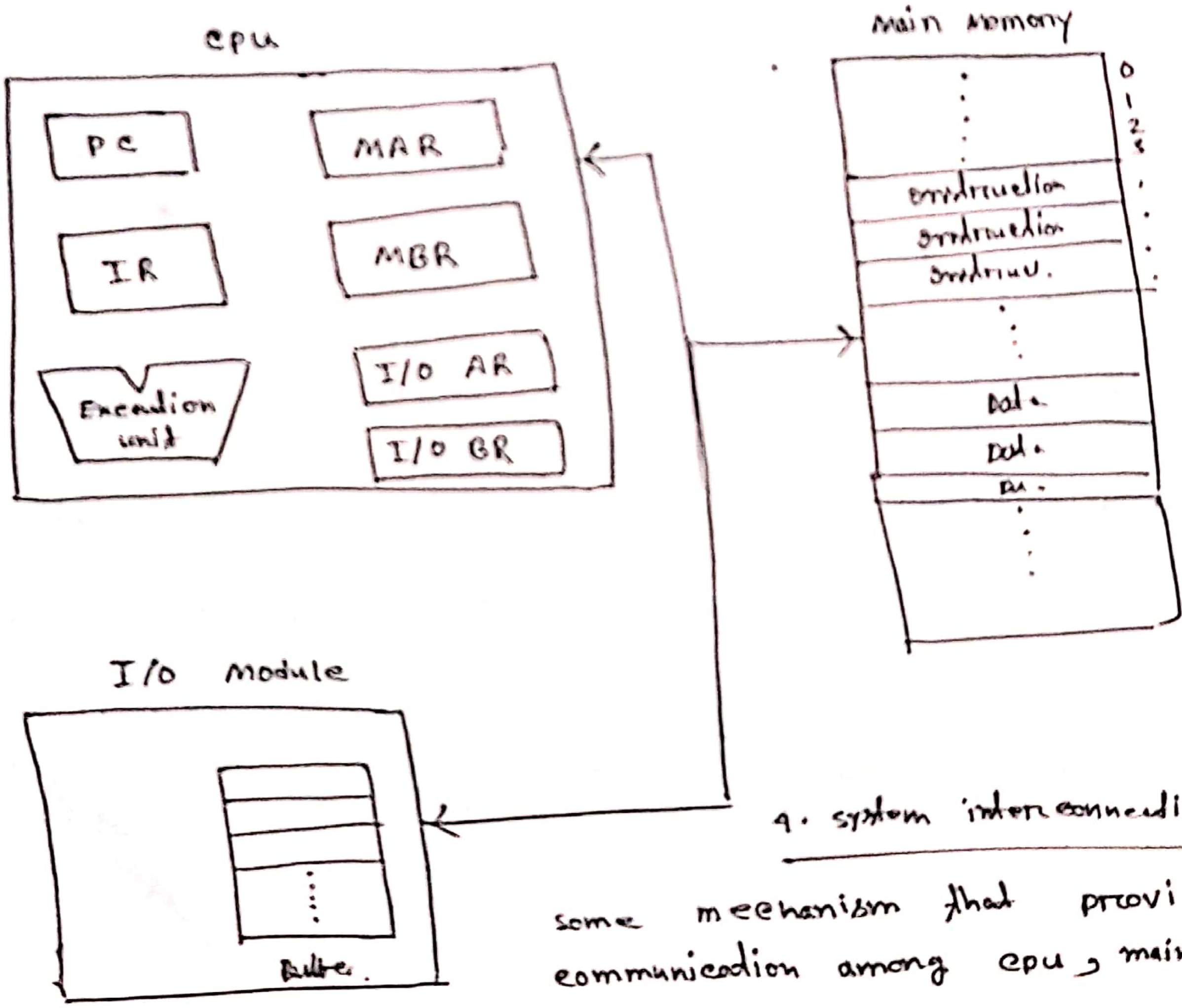


1. CPU :- Controls the operation and perform data processing function.

2. I/O :- moves data between comput. and external environment.

3. Main Memory : store data

⇒ Computer Components : Top Level View



4. system interconnection :

some mechanism that provides communication among cpu, main memory, and I/O :

⊕ What are the roles of different register of controll unit and ALU ?

Different register of controll unit are :-

1. MAR (Memory Address Register)
2. MBR (Memory Buffer Register)
3. PC (program counter)
4. IR (Instruction Register)
5. Accumulator Register
6. stack controll register
7. Flag register.

MAR :- MAR are used to handle data transfer between the main memory and the processor. MAR holds the address of the main memory to or from which data is to be transferred.

MBR :- main memory and processor. MBR contains data which is going to be read or written.

PC :- A program is a series of instruction stored in the memory. This instruction must be executed in a proper order. This sequence of instruction monitored by PC

IR : IR is used to hold the instruction that is currently being executed.

(E)

ALU :- Arithmetic Logic unit ALU is a digital circuit used to perform arithmatic and logic operations. ALU performs simple addition, subtraction, multiplication, division, and logic operation like AND, OR

ALU represent the fundamental building block of the cpu of a computer. modern computer contains very powerful and complex ALU.

Q) How are data read from a magnetic disk ?

⇒ Magnetic disk is a direct access secondary memory or storage device. It is a thin plastic or metalic circular plate coated with magnetic oxide and encased in a protective cover. Data is stored on magnetic disk as magnetized spots. The presence of magnetic spot represent 1 bit and its absence represent bit 0.

As the disk is spinning, a read or write head moves accross its surface. To read data the head makes a note of whether the section is magnetised or not. To write data the head magnetise or demagnetise a section of the disk.

Q:- Distinguish between DRAM and SRAM of characteristics such as application, speed, size.

Characteristic	SRAM	DRAM
speed	Faster	Slower
cost	Expensive	cheap
size	small	Large
used in	cache memory	main memory
power consumption	High	Low

(ii):- Mention the concept of von Neumann Architecture:-

The von Neumann architecture also known as the von Neuman model is a computer architecture based on 1945 description by John von Neumann. It includes with these components:

- A processing unit with both an arithmetic-Logic unit and processor register.
- A control unit that includes an instruction register and a program counter.
- Memory that stores data and instruction.
- External mass storage.
- Input and output mechanism.

Q:- Difference between system Bus and Local Bus :-

Local Bus :- In computer architecture, a local bus is a computer bus that connects directly from the central processing unit (cpu) to one or more slots on the expansion Bus.

- Local Bus helps cpu to avoid bottleneck created by the expansion Bus.
- Local Bus increase the speed of data transfer.
- Example of local Bus :- VESA Local Bus
  Processor Direct slot

System Bus :- The system Bus is a pathway between a computer Microprocessor and the main memory.

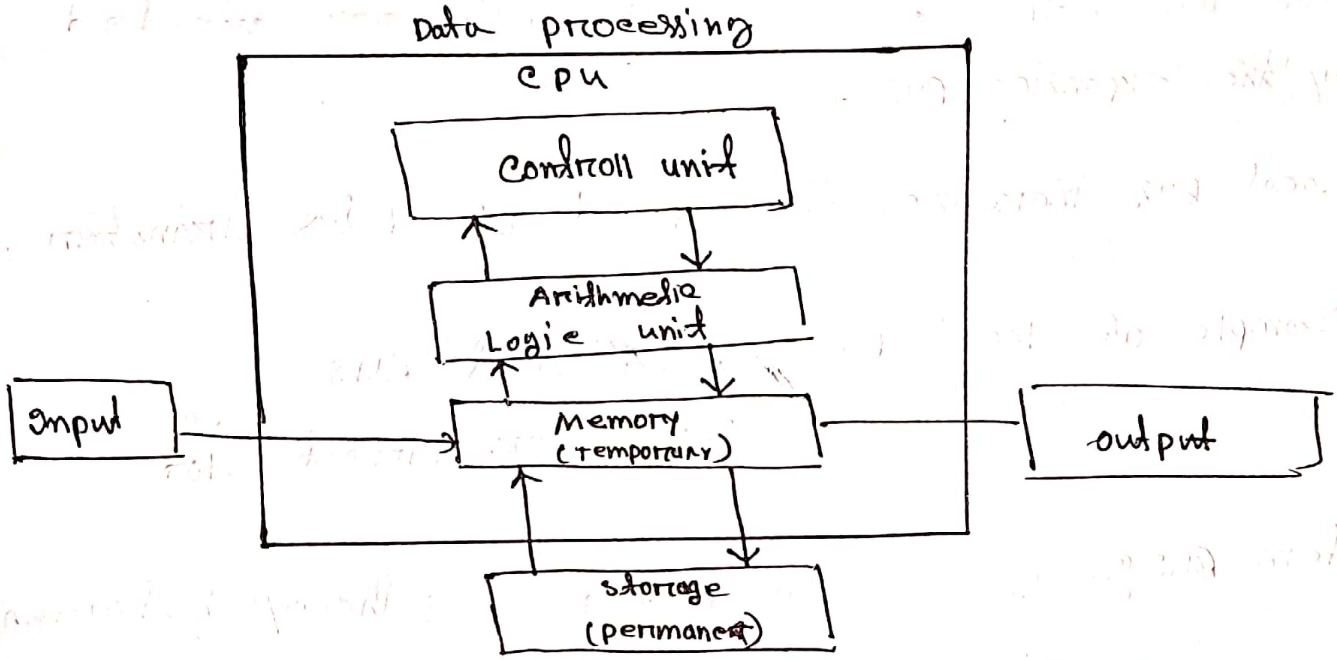
- It provides communication path for the data and control signal moving between major components of computer system.

- system bus combines of three main bus: namely bus, ~~Ad~~ Data Bus, control Bus, Address Bus.

(A) Explain the basic function of a computer :

There are four basic function of the computer :

1. Data input
2. Data processing
3. information output
4. Data storage.



Computer Architecture Vs Computer Organization :-

Arch	
* computer architecture define how hardware component connected together and form a computer.	* Computer organization define the structure and behavior of a computer system seen by the user.
* It acts as the interface between hardware and software.	* It deals with the component of a connection in a system
* Computer architecture helps us to understand the functionalities of a system.	* Organization tell us how all the units are interconnected
computer architecture deals with high Level design issues.	Low level design issues.

Explain why we study computer architecture and organization :-

what are the basic of Von Neuman architecture :-

List and explain the key characteristic of computer family :-

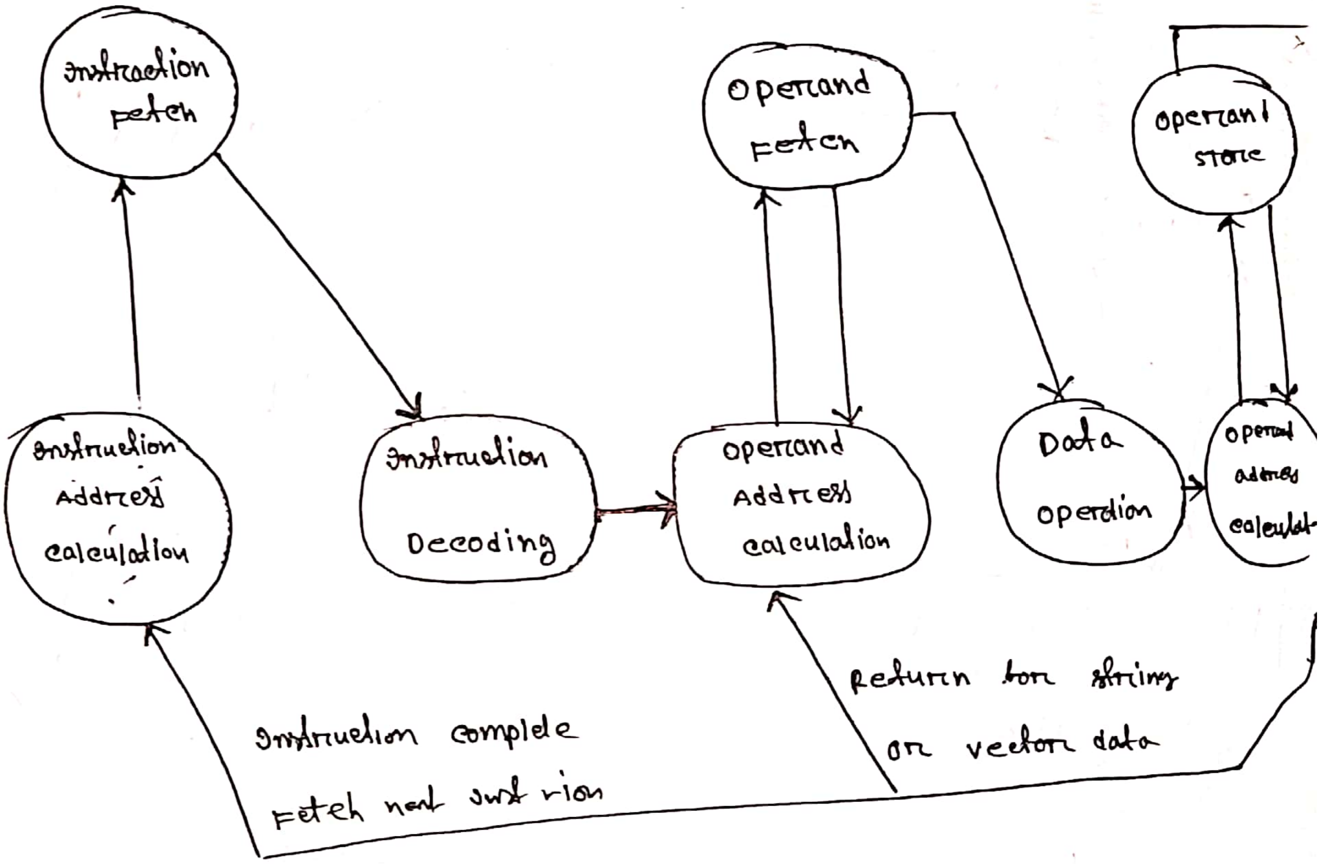
Sometimes / In many cases, the same set of machine instruction is supported on all the members of the family. Thus a program that execute on one machine will also execute on any other.

There are some characteristics of a computer family :

1. Similar or identical operating system.
2. Increasing speed
3. Increasing Number of I/O ports
4. Increasing memory size
5. Increasing cost.

(b) Explain the possible states that define an instruction execution :-

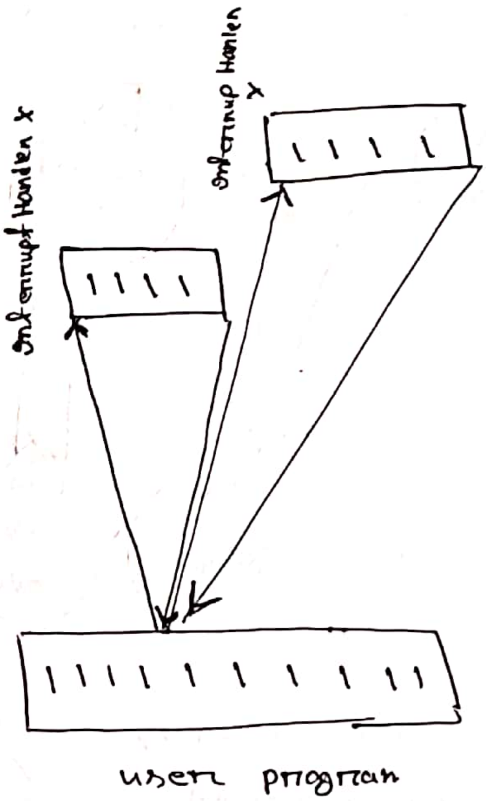
1. instruction address calculation
2. instruction fetch
3. instruction decoding
4. operand address calculation.
5. operand fetch,
6. Data operation
7. operand store



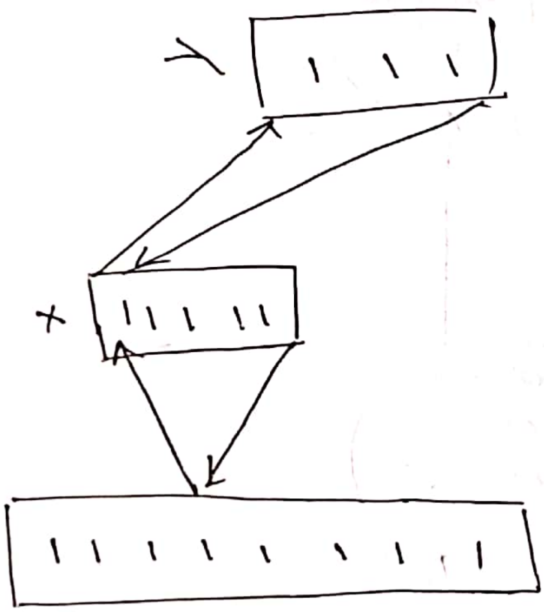
List and Briefly define two approaches to dealing with multiple interrupts :-

First Approach to dealing with multiple interrupts :-

The first approach is to disable interrupts while an interrupt is being processed. It means when a user program is executing and an interrupt occurs, interrupts are disabled immediately. After the user program execution done the interrupt enabled.



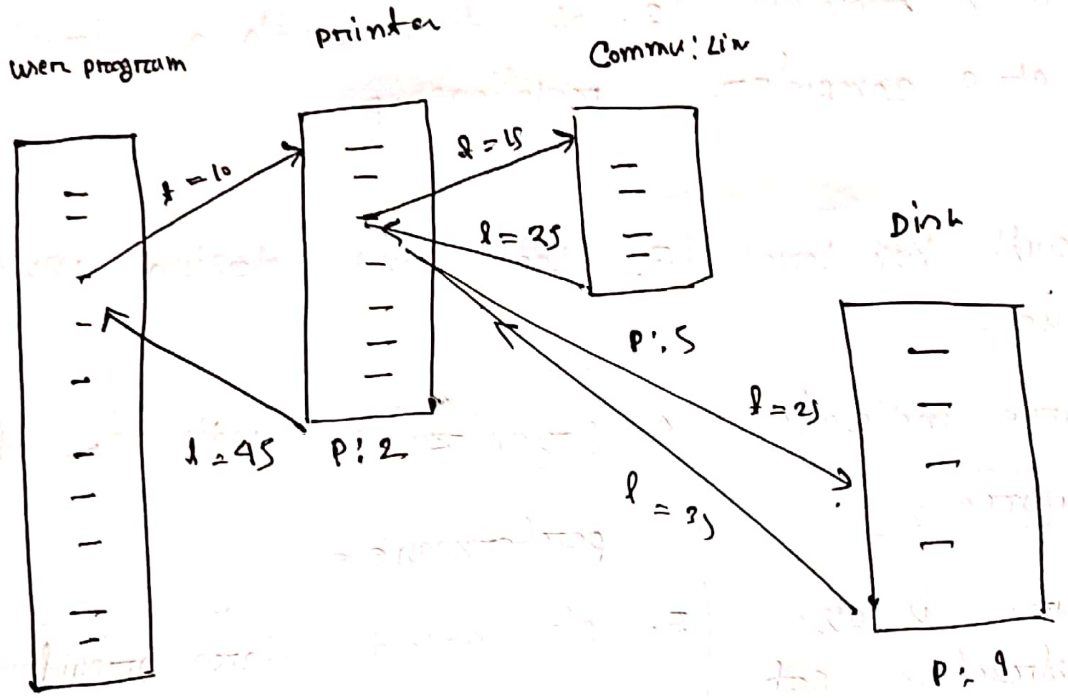
sequential interruption



Nested interruption.

second Approach to dealing with multiple interrupts :-

The second approach is to define priorities for interrupts to allow interrupt of higher priority.



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Computer Architecture	Organization
1. Architecture describes what the computer does.	2. Organization describes how it does it.
2. It deals with functional behavior of a computer system.	2. It deals with a structural relationship.
3. It deals with high level design issue.	3. Low Level design issue.
4. Architecture indicates its hardware.	4. Organization indicates its performance.
5. Architecture is also called Instruction set architecture (ISA)	5. Its called micro architecture.

Computer structure Vs Function :-

Computer structure	Function
1. The way in which the component are inter-related.	1. The operation of each individual component as a part of structure.
2. There are four main structure.	2. There are four main function.
3. Lower Layer of hierchy	3. Lower Layer of hierchy.

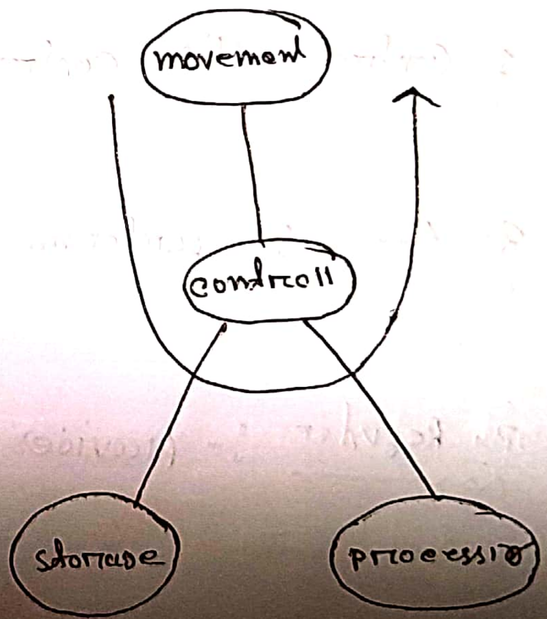
CO

# Types of structure :	# Types of function :
1. CPU	1. Data processing
2. Main memory	2. Data storage
3. I/O	3. Data Momement
4. System interconnection / Bus	4. Control

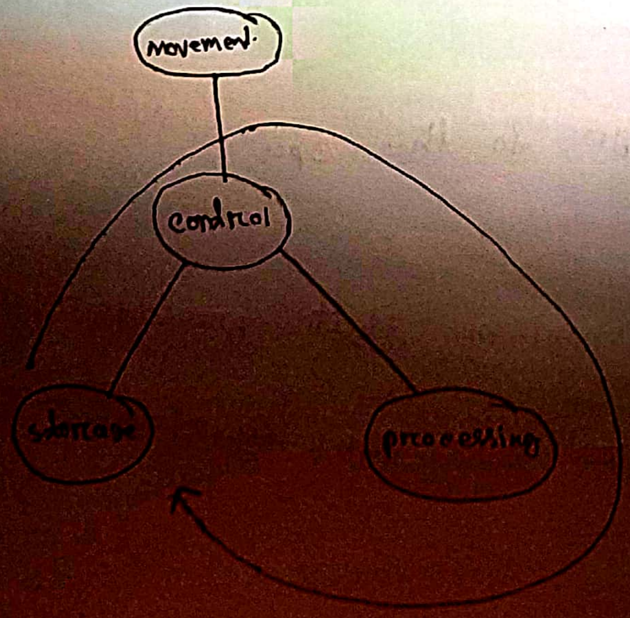
Main four function of computer :-

1. Data processing
2. Data storage
3. Data movement
4. controll

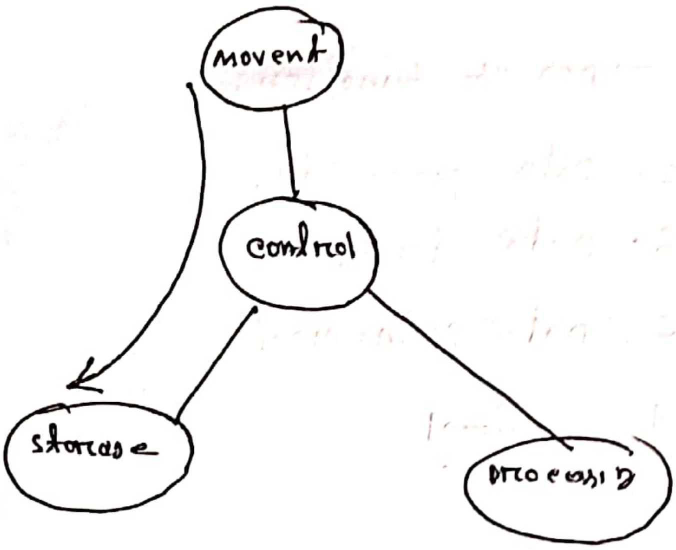
Data movement :



Data processing



storage :



(#) List and briefly define the main structure component of a processor :-

1. Control unit : Controls the operation of cpu.

2. ALU :- performs data processing function

3. Register :- provides storage to the cpu

4. cpu interconnection :- mechanism that provides communication among control unit, ALU, Register.

Stored program Computer :- A computer that stores instruction in its memory to perform variety of task sequentially.

Moors Law : The moors law predicted that the number of transistor in a dense integrated Circuit doubles every two years and the price would be half.

key distinguishing feature of MP :-

CPU vs Core :-

Core	CPU
(i) Core is an execution unit inside CPU	(i) CPU is an electronic circuit that carries out instruction.
(ii) Receive and execute instruction	(ii) Carries out instruction to perform : 1. arithmetic operation 2. control 3. I/O operation.

Two approaches to develop an embeded operating system:

⇒ The first approaches is to take an existing OS and adapt it for the existing system. For example embeded version of Linux, windows.

The other approaches is to design and implement an OS.

Version of cortex M-series:

1. Cortex-M0: 8, 16 bit
2. Cortex-M0+: 8, 16 bit is more energy efficient.
3. Cortex-M3: for 16 and 32 bit
4. cortex-M4: Enhance version of M3 and energy efficient.

Chapter : 2

(2.1) Define some of the technique used in processor for increasing speed.

1. Pipelining : Transfer data and instruction simultaneously and fetch and execute at a time.

2. Branch prediction : processor try to guess which way a branch or group of instruction will go.

3. Super scalar Execution : Ability to issue more than one instruction in every clock cycle.

4. Data flow Analysis : processor analyse which instrs are dependent on each other.

(2.2) performance Balance :-

Adjusting the organization and architecture to make up for the downside of various component.

(2.3) Difference among multicore system, MICs and GPGPUs